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Method of manufacturing a semiconductor device

Abstract

A method of manufacturing a semiconductor device, including: forming a dielectric layer configured to be a gate oxide contacting the second well on the substrate, wherein the dielectric layer is single-layered dielectric layer and includes a contact via penetrating through the dielectric layer; and forming a patterned conductive layer contacting the dielectric layer, wherein the patterned conductive layer includes a first conductive portion isolated from the second well and configured to be a gate electrode, and a second conductive portion coupled to the first well via the contact via; wherein the first conductive portion is leveled with the second conductive portion, and the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the dielectric layer and the first conductive portion collectively serve as a gate of the transistor, and the transistor is configured as a high-voltage transistor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/205,797, field on Nov. 30 2018, which is a divisional of U.S. application Ser. No. 15/061,596, filed on Mar. 4, 2016, which claims priority to U.S. Provisional Application No. 62/288,806, filed on Jan. 29, 2016, which application is hereby incorporated herein by reference.

BACKGROUND
(1) High-voltage metal-oxide-semiconductor (HVMOS) devices are widely used in many electrical devices, such as input/output (I/O) circuits, central processing unit (CPU) power supplies, power management systems, and alternating current/direct current (AC/DC) converters. There are a variety of forms of HVMOS devices. A symmetric HVMOS device may have a symmetric structure on the source side and drain side. High voltage can be applied on both drain and source sides. In contrast, an asymmetric HVMOS device may have asymmetric structures on the source side and drain side.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1**A is a cross-sectional view of a transistor of a semiconductor device, in accordance with some embodiments of the present disclosure.

(3) FIG. **1**B is a layout view of the transistor shown in FIG. **1**A, in accordance with some embodiments of the present disclosure.

(4) FIGS. **1**C to **1**F are diagrams showing a method of manufacturing the transistor of the semiconductor device shown in FIG. **1**A, in accordance with some embodiments of the present disclosure.

(5) FIG. **2**A is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

(6) FIG. **2**B is a layout view of the semiconductor device shown in FIG. **2**, in accordance with some embodiments of the present disclosure.

(7) FIG. **3**A is a cross-sectional view of a semiconductor device including a first transistor and a second transistor, in accordance with some embodiments of the present disclosure.

(8) FIGS. **3**B to **3**G are diagrams showing a method of manufacturing the semiconductor device shown in FIG. **3**A, in accordance with some embodiments of the present disclosure.

(9) FIG. **4** is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

(10) FIG. **5** is a flow chart illustrating a method of forming a semiconductor device, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) FIG. **1**A is a cross-sectional view of a transistor **10** of a semiconductor device **1**, in accordance with some embodiments of the present disclosure. Referring to FIG. **1**A, the transistor **10** includes a gate **11** on a substrate **18**, and an active region **12** in the substrate **18**.

(13) In some embodiments, the substrate **18** includes a semiconductor material such as silicon. In other embodiments, the substrate **18** includes silicon germanium, gallium arsenic, or other suitable semiconductor materials. In some other embodiments, the substrate **18** further includes other features such as a buried layer, and/or an epitaxy layer. The buried layer may be doped with antimony (Sb) via ion implantation to a concentration of about 5.0×10.sup.13 to about 1.5×10.sup.14 at an energy between about 70 keV and about 90 keV, and to a depth of greater than about 2 micrometers. Skilled artisans will recognize that other n-type dopants may be used depending on the design requirements of the device. For example, antimony exhibits less

autodoping during epitaxy and the following heat cycles, but has a lower solubility limit compared to arsenic, which may necessitate higher anneal temperatures to activate the antimony. Furthermore, in some embodiments, the substrate **18** is semiconductor on insulator such as silicon on insulator (SOI). In other embodiments, the semiconductor substrate **18** includes a doped epi layer, a gradient semiconductor layer, and/or further includes a semiconductor layer overlying another semiconductor layer of a different type such as a silicon layer on a silicon germanium layer. In some other examples, a compound semiconductor substrate includes a multilayer silicon structure or a silicon substrate may include a multilayer compound semiconductor structure. In some embodiments, the substrate **18** may include other elementary semiconductors such as germanium and diamond. In some embodiments, the substrate **18** includes a compound semiconductor such as, silicon carbide, gallium arsenide, indium arsenide, or indium phosphide.
(14) The gate **11** includes a patterned conductive layer **14** serving as its gate electrode, and an insulating layer **16** serving as its gate oxide.
(15) The patterned conductive layer **14** is disposed on the insulating layer **16**, which in turn is disposed on the active region **12**, The active region **12** includes a first well **124**, where a channel **126** is defined. The first well **124** of a first dopant type is disposed between a second well **120** and a third well **122** both of a second dopant type opposite to the first dopant type. The second well **120** serves as a first source/drain region of the transistor **10**, and the third well **122** serves as a second source/drain region of the transistor **10**. In the present embodiment, the substrate **18** is a p-type substrate, the first well **124** includes a high voltage n-well (HVNW), and each of the second well **120** and the third well **122** includes a high voltage p-well (HVPW). As a result, the transistor **10** includes a p-type transistor structure. For example, the transistor **10** includes a p-type metal-oxide-semiconductor (PMOS) transistor, or a p-type metal-oxide-semiconductor field effect transistor (PMOSFET). Although only one gate structure is illustrated, it is understood that the transistor **10** may include a number of gate structures for PMOS transistors, including short channel and long channel transistors.
(16) Those skilled in the art will recognize that the embodiments disclosed herein by way of a PMOS transistor may also be applicable to an NMOS transistor, an n-type metal-oxide-semiconductor field effect transistor (NMOSFET), and an NPN transistor. For example, the first well **124** includes a high voltage p-well, and each of the second well **120** and the third well **122** includes a high voltage n-well. Additionally, while the dopants discussed in specific terms and with reference to specific doping materials, concentrations and doping depths, those skilled in the art will recognize that alternative doping characteristics may be advantageously employed.
(17) Furthermore, the patterned conductive layer **14** is, for example, formed in an interconnection layer such as a metal-1 (M1) layer in a semiconductor manufacturing process. Furthermore, since the patterned conductive layer **14** is a portion of the metal-1 layer, unlike some existing transistors, the patterned conductive layer **14** is free of spacers at its sides. Accordingly, the process for manufacturing the transistor **10** in the present embodiment is relatively simple.
(18) In the present embodiment, a layer (such as the patterned conductive layer **14**) serving as a gate electrode is in a metal-1 layer. However, the present disclosure is not limited thereto. In another embodiment, a layer in a metal-2 layer serves as a gate electrode of the transistor **10**.
(19) The insulating layer **16**, serving as another component of the gate **11** of the transistor **10**, fully covers the active region **12**. Moreover, the insulating layer **16** fully covers the substrate **18** as the patterned conductive layer **14** forms a portion of an interconnection layer. The insulating layer **16** is configured to insulate the active region **12** from the patterned conductive layer **14**. Ideally, with the insulating layer **16**, the active region **16** is electrically isolated from the patterned conductive layer **14** if there is not any interconnection features therebetween to couple one to the other.
(20) In an embodiment, the insulating layer **16** includes an interlayer dielectric (ILD) layer. The insulating layer **16** includes doped silicon glass such as phosphorous silicon glass (PSG) or boron phosphorous silicon glass (BPSG). In some embodiments, the insulating layer **16** includes silicon

oxide, silicon nitride, silicon oxynitride, spin-on glass (SOG), fluorinated silica glass (FSG), carbon doped silicon oxide (e.g., SiCOH), BLACK DIAMOND® (Applied Materials of Santa Clara, Calif.), XEROGEL®, AEROGEL®, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), FLARE®, SILK® (Dow Chemical, Midland, Mich.), polyimide, other proper porous polymeric materials, other suitable dielectric materials, and/or combinations thereof. In some embodiments, the insulating layer **16** includes a high density plasma (HDP) dielectric material (e.g., HDP oxide) and/or a high aspect ratio process (HARP) dielectric material (e.g., HARP oxide).

(21) The insulating layer **16** has a thickness T**2**, and the patterned conductive layer **14** has a thickness T**1**. In an embodiment, the thickness T**2** of the insulating layer **16** is about 1200 angstrom (Å). It is understood that the insulating layer **16** may include one or more dielectric materials and/or one or more dielectric layers.

(22) In some existing transistors, a gate structure of a transistor includes a gate oxide, and a poly serving as a gate electrode. A total thickness of the gate structure is ideally the sum of the thickness of the gate oxide and poly. Typically, a dielectric layer is adapted to cover the gate and an active region of the transistor, such that the gate and the active region are insulated from a metal layer (such as a metal-1 layer) for routing. In order to cover the gate of the transistor, the thickness of the gate is not allowed to be thicker than that of the dielectric layer. Otherwise, the gate cannot be covered by the dielectric layer, and consequently the gate of the transistor may physically contact a patterned conductive layer for routing. The transistor may accordingly work abnormally and even be damaged.

(23) With the continuous development in semiconductor manufacturing, dimensions or features of a semiconductor device become increasingly smaller. As a result, the thickness of a dielectric layer in the semiconductor device also shrinks as well. However, in some applications, such as power supply systems, a semiconductor device is asked for enduring a relatively high voltage. Taking such a transistor for instance, to sustain the relatively high voltage, thickness of a gate oxide of the transistor would become relatively thicker, which is against the trend of downsizing in semiconductor manufacturing. Nevertheless, in such applications, the gate is inevitably thicker than the dielectric layer.

(24) In an embodiment, in a high voltage application, the thickness of a gate oxide of a transistor for enduring the high voltage is approximately 800 angstrom. Furthermore, the thickness of a poly of the transistor is approximately 800 angstrom. Therefore, a total thickness of the gate on the substrate surface is approximately 1600 angstrom. For example, in the 28-nm process technology, the design rule limits the thickness of a dielectric layer measured from a substrate surface to be no more than approximately 1200 angstrom. In that case, the gate is thicker than the dielectric layer. As a result, the dielectric layer cannot cover the gate of the transistor, and accordingly cannot insulate the gate of the transistor from a patterned metal layer for routing. It is therefore desirable to have a transistor structure according to the present disclosure to meet the size requirements for the advanced processes (such as the 28-nm process technology).

(25) In the present disclosure, the insulating layer **16** (i.e., the dielectric layer) is taken as a gate oxide of the gate **11** (i.e., as a part of the transistor **10**). Such arrangement overcomes the issues of size limitation encountered by the existing transistors. In the advanced processes, the insulating layer **16** may has a thickness smaller than 1200 angstrom (Å). Moreover, since the insulating layer **16** is typically thicker than the gate oxide of the existing transistors, the transistor **10** according to the present embodiment is able to sustain a higher voltage than the existing transistors.

(26) Furthermore, in the existing transistors, a poly serving as a gate electrode is independent of a metal-1 layer for routing. Contrarily, in the present embodiment, the patterned conductive **14** serving as a gate electrode is a portion of an interconnection layer (such as a metal-1 layer) for routing. Taking the patterned conductive layer **14** out of the metal-1 layer for a component of the gate **11** does not make the manufacturing process complex. Besides, no additional cost is incurred.

Furthermore, since the patterned conductive layer **14** takes advantage of the metal-1 layer, no spacer is required at its sides. Effectively, the manufacturing process is simplified.
(27) In a high voltage device, an off-type breakdown voltage refers to a breakdown voltage measured under the condition that a gate of a transistor receives a reference ground voltage. Moreover, an on-type breakdown voltage refers to a breakdown voltage measured under the condition that a gate of a transistor receives a logic high voltage. The two breakdown voltages are important performance indicators of a power transistor. Typically, the on-and-off-type breakdown voltages are determined by the design of an active region (such as the first well **124**, the second well **120** and the third well **122** in the present embodiment) of the transistor, and are not relative to an arrangement of a gate of the transistor. The design involves an arrangement of wells in the active region, and the associated concentration, depth and width thereof. The present disclosure overcomes the aforesaid issues due to size limitation in advanced processes without modifying (or changing) the design of the first well **124**, the second well **120** and the third well **122**. Therefore, the on-and-off-type breakdown voltages are not affected. Performance relative to the on-and-off-type breakdown voltages are kept substantially the same.
(28) The present disclosure is applicable to other semiconductor devices, where the gate of a transistor can be made thicker than a dielectric layer.
(29) FIG. **1**B is a layout view of the transistor **10** shown in FIG. **1**A, in accordance with some embodiments of the present disclosure. Referring to FIG. **1**B, the insulating layer **16**, serving as a gate oxide of the transistor **10**, fully covers the substrate **18**. The patterned conductive layer **14** is on the insulating layer **16**, and over the second well **120** and the third well **122** in the substrate **18**. Based on such layout design, for the similar reasons as those discussed and illustrated with reference to FIG. **1**A, the issue that the gate of the existing transistors is thicker than a dielectric layer is prevented.
(30) FIGS. **1**C to **1**F are diagrams showing a method of manufacturing the transistor **10** of the semiconductor device **1** shown in FIG. **1**A, in accordance with some embodiments of the present disclosure. It is understood that FIGS. **1**C to FIG. **1**F have been simplified for clarity. Referring to FIG. **1**C, a substrate **18** is received. In some embodiments, the substrate **18** includes a p-type substrate.
(31) Referring to FIG. **1**D, a first well **124**, a second well **120** and a third well **122** are defined in the substrate **18** by, for example, a masking process and one or more ion implantation process followed by a drive-in processes. In the present embodiment, the first well **124** serves as a high voltage p-well (PVNW), and each of the second well **120** and the third well **122** serves as a high voltage n-well (HVNW). The first well **124** can be defined firstly, and then the second well **120** and the third well **122** are subsequently defined. Skilled practitioners will recognize that the order of implantation of the first well **124**, the second well **120** and the third well **122** may be varied without deviating from the spirit of the presented disclosure.
(32) In order to provide a vertically controlled profile in a doped region such as each of the first well **124**, the second well **120** and the third well **122**, multiple successive implants may be performed to create multiple implant regions. The successive implants may be used to tailor the doping profile by varying the implant energy, concentration and depth of each implant step. Additionally, the successive implant steps may implant different dopants to further customize the doping profile for a particular doped region.
(33) Referring to FIG. **1**E, an insulating layer **16** is formed on the substrate **18** by, for example, a deposition process followed by a planarization process.
(34) Referring to FIG. **1**F, a patterned conductive layer **14** is formed on the substrate **16** and over the first well **124**, the second well **120** and the third well **126** by, for example, a deposition process for depositing a metal-1 layer, followed by an etching process.
(35) FIG. **2**A is a cross-sectional view of a semiconductor device **2** for explaining routing, in accordance with some embodiments of the present disclosure. Referring to FIG. **2**A, the

semiconductor device **2** is similar to the semiconductor device **1** described and illustrate with reference to FIG. **1**A except that, for example, the semiconductor device **2** includes a transistor **20**, an interconnect feature **26** and an interconnect feature **28**. The transistor **20** is similar to the transistor **10** described and illustrated with reference to FIG. **1**A except that, for example, the transistor **20** includes a gate **21** further including a patterned conductive layer **24**. The interconnect feature **26**, which communicates the interconnect feature **28** to a second source/drain region **122** of the transistor **20**, is configured for routing. Moreover, a heavily doped region (not shown) is formed in the second source/drain region **122** for ohmic contact with the interconnect feature **26**. The interconnect feature **26** is formed as, for example, a contact in a semiconductor manufacturing process. Furthermore, the patterned conductive layer **24** serving as a gate electrode of the transistor **20** and the interconnect feature **28** are in the same interconnection layer such as a metal-1 (M1) layer in a semiconductor manufacturing process.

(36) FIG. **2**B is a layout view of the semiconductor device **2** shown in FIG. **2**, in accordance with some embodiments of the present disclosure. FIG. **2**B is shown to better explain an application of the patterned conductive layer **24** and the interconnect feature **28** being in the same interconnection layer. Referring FIG. **2**B, the patterned conductive layer **24** and the interconnect feature **28** are in the same interconnection layer, but independent of (or, separated from) each other. The interconnect feature **26** in the interconnect feature **28** communicates to the second source/drain region **122**.

(37) FIG. **3**A is a cross-sectional view of a semiconductor device **3** including a first transistor **30** and a second transistor **31**, in accordance with some embodiments of the present disclosure. Referring to FIG. **3**A, the semiconductor device **3** is similar to the semiconductor device **1** described and illustrated with reference to FIG. **1**A, except that the semiconductor device **3** includes the first transistor **30** operating in a first voltage domain and the second transistor **31** operating in a second voltage domain different from the first voltage domain. Specifically, the first transistor **30** operates in a relatively high voltage domain (such as, 3.3 V), and therefore is called a high voltage (HV) transistor. Contrarily, the second transistor **31** operates in a relatively low voltage domain (such as, 1.1 V or 1.2 V), and therefore is called a low voltage (LV) transistor. However, the second transistor **31** is not limited thereto, and may operate in a voltage domain other than the relatively low voltage domain. For example, the second transistor **31** operates in a normal voltage domain of 2.5 V. In that case, the second transistor **31** is called a normal voltage transistor.

(38) The first transistor **30** is similar to the transistor **10** described and illustrated with reference to FIG. **1**A except an insulating layer **32**. Like the insulating layer **16** described and illustrated with reference to FIG. **1**A, the insulating layer **32** also fully covers the substrate **18**. Accordingly, the insulating layer **32** not only covers the active region **12** of the first transistor **30**, but also an active region **32** of the second transistor **31**. Moreover, the insulating layer **32** of a gate **302** of the first transistor **30** encapsulates the gate **316** of the second transistor **31**. The insulating layer **32** has a thickness of T**4**, larger than a thickness T**3** of the gate **316** of the second transistor **31**. In some embodiments, the thickness of T**4** is about 1200 angstrom (Å).

(39) The active region **12** of the first transistor **30** is defined by a plurality of isolation structures **34**. The isolation structure **34** may be filled with an insulator or dielectric material. In an embodiment, the isolation structure **34** includes a shallow trench isolation (STI). Alternatively, the isolation structure **34** includes a local oxidation of silicon (LOCOS) configuration. The isolation structure **34** includes silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate (FSG), and/or a low k dielectric material known in the art.

(40) Moreover, the isolation structure **34** further defines the boundary of wells **36**. The well **36** serves as a pick-up and is configured for electrical connection of the substrate **18**. A voltage can be applied via the well **36** to the substrate **18**.)

(41) The second transistor **31** includes the gate **316** on the substrate **18** and the active region **32** in the substrate **18**. The active region **32** includes a well **318**, a first doped region **311** and a second

doped region **313** in the well **318**. Moreover, the well **318** is isolated from the well **36** by a well **38**. The well **38** has a dopant type opposite to the well **318** and the well **36**.
(42) The well **318** includes a first dopant type, and the first doped region **311** and the second doped region **313** include a second dopant type opposite to the first dopant type. In the present embodiment, the well **318** is a p-well, the first doped region **311** and the second doped region **313** are n-type doped regions. As a result, the second transistor **31** includes an n-type transistor structure. The second transistor **31**, for example, includes an n-type metal-oxide-semiconductor (NMOS), or an n-type metal-oxide-semiconductor field effect transistor (NMOSFET). Although only one gate structure is illustrated, it is understood that the transistor **31** may include a number of gate structures for NMOSs including short channel and long channel transistors.
(43) Those skilled in the art will recognize that the embodiments disclosed herein by way of an NMOS transistor may also be applicable to a PMOS transistor, a p-type metal-oxide-semiconductor field effect transistor (PMOSFET), a PNP transistor, or the like. For example, the well **318** is an n-well, and the first doped region **311** and the second doped region **313** are p-type doped regions. Additionally, while the dopants discussed in specific terms and with reference to specific doping materials, concentrations and doping depths, those skilled in the art will recognize that alternative doping characteristics may be advantageously employed.
(44) The gate **316** includes a gate oxide **314** on the substrate **18**, and a poly **312**, serving as a gate electrode, on the gate oxide **314**. The gate **316** includes spacers at its sides, but for clarity of illustration, the spacers are omitted herein. Unlike some existing transistors designed for operating under a relatively high voltage, the second transistor **31** operates under a relatively low voltage, and a thickness of the gate oxide **314** is therefore relatively thin. Therefore, the thickness T**3** of the gate **316** does not exceed that of a dielectric layer (i.e., in the present embodiment, as shown in FIG. **3**A, the thickness T**3** is thinner than the thickness T**4**), wherein the dielectric layer is adapted to insulate a gate from a patterned conductive layer for routing. The gate **316** of the second transistor **31** does not encounter the problems in the existing transistors designed for operating under a relatively high voltage.
(45) In the present embodiment, for the second transistor **31**, the insulating layer **32** serves as a dielectric layer for insulating the gate **316** and active region **32** from a patterned conductive layer (not shown) for routing. In contrast, for the first transistor **30**, the insulating layer **32** serves as a gate oxide for the gate **302**.
(46) With an insulating layer serving as a gate oxide of a high voltage transistor and serving as a dielectric layer for a low or normal voltage transistor, the high voltage transistor and the low or normal voltage transistor can be integrated easily without complicating the semiconductor manufacturing process. Specifically, since the insulating layer **32** is taken as a gate oxide of the first transistor **30**, a photolithography process is simplified and at least three masks are eliminated.
(47) FIGS. **3**B to **3**G are diagrams showing a method of manufacturing the semiconductor device **3** shown in FIG. **3**A, in accordance with some embodiments of the present disclosure. Referring to FIG. **3**B, a substrate **18** is received. In some embodiments, the substrate **18** includes a p-type dopant.
(48) Referring to FIG. **3**C, isolation structures **34** are formed in the substrate **18** by, for example, a deposition process, an etching process, a pullback process, an annealing process and a chemical mechanical planarization process sequentially performed in order. In an embodiment, the isolation structures **34** include an STI structure.
(49) Referring to FIG. **3**D, a first well **124**, a second well **120**, a third well **122**, a well **36**, wells **38** and a well **318** are defined in the substrate **18** by, for example, an ion implantation process followed by a drive-in process. In some embodiments, the first well **124** serves as a high voltage p-well (HVPW), the second well **120** serves as a high voltage n-well (HVNW), the third well **122** also serves as high voltage n-well, the wells **36** serve as a high voltage p-well and the well **38** serves as a high voltage n-well.

(50) Afterwards, a first doped region **311** and a second doped region **313** are defined in the well **318** by, for example, an ion implantation process followed by a drive-in process. In some embodiments, the first doped region **311** and the second doped region **313** serve as n-type doped regions.

(51) Referring to FIG. **3**E, a gate oxide **314** is formed on the substrate **18** by, for example, a deposition process followed by a photolithography process. Afterwards, a poly **312** is formed on the gate oxide **314** by, for example, a deposition process followed by a photolithography process.

(52) Referring to FIG. **3**F, an insulating layer **32** is formed on the substrate **18** and the poly **312**, by, for example, a deposition process followed by a planarization process such as a chemical mechanical polishing (CMP).

(53) Referring to FIG. **3**G, a patterned conductive layer **14** is formed on the insulate layer **32** over the first well **124** between the second well **120** and the third well **122** by, for example, a deposition process followed by an etching process.

(54) FIG. **4** is a cross-sectional view of a semiconductor device **4**, in accordance with some embodiments of the present disclosure. Referring to FIG. **4**, the semiconductor device **4** is similar to the semiconductor device **3** described and illustrated with reference to FIG. **3**A except that, for example, the semiconductor device **4** further includes a patterned conductive layer **42** for routing of a second transistor **31**. Moreover, the semiconductor device **4** includes an interconnect feature **44**, in an insulating layer **32**, configured to connect a gate **316** of the second transistor **31** to the patterned conductive layer **42**. In the present embodiment, the patterned conductive layer **42** for routing of the second transistor **31** and a patterned conductive layer **14** for routing of the first transistor **30** are in the same conductive layer (or the same interconnection layer).

(55) As discussed in the embodiment of FIG. **3**A, in the semiconductor device **4**, with the insulating layer **32** serving as a gate oxide of a high voltage transistor **30** and serving as a dielectric layer for a low or normal voltage transistor **31**, the high voltage transistor **30** and the low or normal voltage transistor can be integrated together without complicating the semiconductor manufacturing process.

(56) FIG. **5** is a flow chart illustrating a method **5** of forming a semiconductor device, in accordance with some embodiments of the present disclosure. Referring to FIG. **5**, in operation **50**, a substrate is provided. The substrate is similar to the substrate **18** described and illustrated with reference to FIG. **3**A.

(57) In operation **52**, a first active region associated with a first transistor is defined in the substrate, and a second active region associated with a second transistor is also defined in the substrate. The first active region associated with the first transistor is similar to the active region **12** described and illustrated with reference to FIG. **3**A. Additionally, the second active region associated with the second transistor is similar to the active region **32** described and illustrated with reference to FIG. **3**A.

(58) In operation **54**, a second gate associated with the second transistor is formed on the second active region associated with the second transistor. The second gate and the second active region define the second transistor. The second gate associated with the second transistor is similar to the gate **316** described and illustrated with reference to FIG. **3**A.

(59) In operation **56**, an insulating layer covering the second gate, the second active region, the first active region and the substrate is formed. The insulating layer serves as a component of a first gate of a first transistor associated with the first active region. The insulating layer is similar to the insulating layer **32** described and illustrated with reference to FIG. **3**A.

(60) In operation **58**, a patterned conductive layer serving as another component of the first gate of the first transistor is formed on the insulating layer. The first gate and the first active region define the first transistor. The patterned conductive layer is similar to the insulating layer **14** described and illustrated with reference to FIG. **3**A.

(61) In some embodiments of the present disclosure, a method of manufacturing a semiconductor

device is disclosed. The method includes forming a dielectric layer configured to be a gate oxide contacting the second well on the substrate, wherein the dielectric layer is single-layered dielectric layer and includes a contact via penetrating through the dielectric layer; and forming a patterned conductive layer contacting the dielectric layer, wherein the patterned conductive layer includes a first conductive portion isolated from the second well and configured to be a gate electrode, and a second conductive portion coupled to the first well via the contact via; wherein the first conductive portion is leveled with the second conductive portion, and the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the dielectric layer and the first conductive portion collectively serve as a gate of the transistor, and the transistor is configured as a high-voltage transistor.

(62) In some embodiments of the present disclosure, a method of manufacturing a semiconductor device is disclosed. The method includes forming an active region including a first well and a second well in a substrate; forming a dielectric layer configured to be a gate oxide on the substrate, wherein the dielectric layer contacts the second well and a top surface of the dielectric layer is horizontal, and the dielectric layer includes a contact via penetrating through the dielectric layer; forming a patterned conductive layer contacting the dielectric layer, wherein the patterned conductive layer includes a first conductive portion and a second conductive portion independent of each other, and the first conductive portion is configured to be a gate electrode and the second conductive portion is coupled to the first well via the contact via; wherein a bottommost surface of the first conductive portion is leveled with a bottommost surface of the second conductive portion, and the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the dielectric layer and the first conductive portion collectively serve as a gate of the transistor, and the interconnect feature is formed before the patterned conductive layer, and the transistor is configured as a high-voltage transistor.

(63) In some embodiments of the present disclosure, a method of manufacturing a semiconductor device is disclosed. The method includes forming an active region including a first well and a second well in a substrate; defining a channel of a transistor in the second well; forming a dielectric layer configured to be a gate oxide on the substrate, wherein the dielectric layer includes a contact via coupled to the first well; and forming a patterned conductive layer on the dielectric layer, wherein the patterned conductive layer includes a first conductive portion configured to be a gate electrode and aligning to the second well, and a second conductive portion coupled to the first well via the contact via, and the second conductive portion is leveled with the first conductive portion; wherein the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the transistor is configured as a high-voltage transistor.

(64) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of manufacturing a semiconductor device, comprising: forming an active region including a first well and a second well in a substrate; forming a dielectric layer configured to be a gate oxide contacting the second well on the substrate, wherein the dielectric layer is single-layered dielectric layer and includes a contact via penetrating through the dielectric layer; and forming a

patterned conductive layer contacting the dielectric layer, wherein the patterned conductive layer includes a first conductive portion isolated from the second well and configured to be a gate electrode, and a second conductive portion coupled to the first well via the contact via; wherein the first conductive portion is leveled with the second conductive portion, and the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the dielectric layer and the first conductive portion collectively serve as a gate of a transistor, and the transistor is configured as a high-voltage transistor.
2. The method of claim 1, wherein the formation of the dielectric layer comprises: forming the dielectric layer fully covering the active region.
3. The method of claim 1, wherein the formation of the dielectric layer comprises: forming the dielectric layer having a thickness of about 1200 angstrom.
4. The method of claim 1, further comprising: defining a channel of the transistor in the second well.
5. The method of claim 1, further comprising: forming the contact via by aligning the contact via to the first well.
6. The method of claim 1, wherein a projection of the second conductive portion in a vertical direction is entirely within the first well.
7. The method of claim 1, wherein the formation of the dielectric layer comprises: forming the dielectric layer across the active region.
8. The method of claim 1, wherein the first conductive portion is disposed over the second well.
9. The method of claim 1, wherein a top surface of the dielectric layer is leveled with a horizontal plane.
10. A method of manufacturing a semiconductor device, comprising: forming an active region including a first well and a second well in a substrate; forming a dielectric layer configured to be a gate oxide on the substrate, wherein the dielectric layer contacts the second well and a top surface of the dielectric layer is horizontal, and the dielectric layer includes a contact via penetrating through the dielectric layer; forming a patterned conductive layer contacting the dielectric layer, wherein the patterned conductive layer includes a first conductive portion and a second conductive portion independent of each other, and the first conductive portion is configured to be a gate electrode and the second conductive portion is coupled to the first well via the contact via; wherein a bottommost surface of the first conductive portion is leveled with a bottommost surface of the second conductive portion, and the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the dielectric layer and the first conductive portion collectively serve as a gate of a transistor, and the transistor is configured as a high-voltage transistor.
11. The method of claim 10, wherein the formation of the patterned conductive layer comprises: forming the patterned conductive layer having the first conductive portion isolated from the second well, and the second conductive portion coupled to the second well.
12. The method of claim 10, wherein the formation of the dielectric layer comprises: forming the dielectric layer fully covering the active region.
13. The method of claim 10, wherein the formation of the dielectric layer comprises: forming the dielectric layer having a thickness of about 1200 angstrom.
14. The method of claim 10, wherein the first conductive portion is disposed over the second well.
15. A method of manufacturing a semiconductor device, comprising: forming an active region including a first well and a second well in a substrate; defining a channel of a transistor in the second well; forming a dielectric layer configured to be a gate oxide on the substrate, wherein the dielectric layer includes a contact via coupled to the first well; and forming a patterned conductive layer on the dielectric layer, wherein the patterned conductive layer includes a first conductive portion configured to be a gate electrode and aligning to the second well, and a second conductive portion coupled to the first well via the contact via, and the second conductive portion is leveled

with the first conductive portion; wherein the first conductive portion and the second conductive portion are formed entirely on a topmost surface of the dielectric layer; wherein the transistor is configured as a high-voltage transistor.

16. The method of claim 15, further comprising: forming the first well and the second well interfacing each other without forming an isolation structure between the first well and the second well.

17. The method of claim 15, further comprising: forming the contact via by aligning the contact via to the first well.

18. The method of claim 17, wherein the formation of the dielectric layer comprises: forming the dielectric layer on the substrate fully covering the substrate.

19. The method of claim 15, wherein the dielectric layer and the first conductive portion collectively serve as a gate of the transistor.

20. The method of claim 15, wherein the first conductive portion is disposed over the second well.
